

# Supply Chain Management

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# Ch 9. Sales and Operations Planning in a Supply Chain

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# Chapter Overview

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1. Responding to Predictable Variability in the Supply Chain
2. Sales and Operations Planning at Red Tomato



## 9.1. Responding to Predictable Variability in the Supply Chain

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- **Predictable variability** is change in demand that can be forecasted
  - Reasons: Seasonal factors, non-seasonal factors
- It can cause high stockouts during peak periods and excessive inventory during periods of low demand, which result in decreased responsiveness and increased costs in the supply chain
- A firm can handle predictable variability through sales and operations planning (S&OP) by appropriately combining two broad options
  - Manage supply using capacity and inventory (**Chap. 8**)
  - Manage demand using short-term price discounts and promotions, thereby spreading demand more evenly (**Chap. 9**)



## 9.1. Responding to Predictable Variability in the Supply Chain

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- Managing supply
  - Managing production capacity: Some approaches to reduce the cost of capacity required to meet predictable variability
    - Time flexibility from workforce
    - Use of seasonal workforce
    - Use of subcontracting
    - Use of dual facilities: specialized and flexible
    - Designing product flexibility into the production processes



## 9.1. Responding to Predictable Variability in the Supply Chain

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- Managing supply
  - Managing inventory: Some approaches to reduce the inventory cost required to meet predictable variability
    - For the products with different peak season demand, use common components across multiple products  
(e.g.) Lawn mowers and snowblowers
    - For the products with the same peak season demand, build products that have more predictable demand during the off-season  
(e.g.) Steady-selling and fashion-trend jumpers



## 9.1. Responding to Predictable Variability in the Supply Chain

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- Managing demand
  - With promotion (or short-term price discounts), a combination of three factors leads to increased demand
    - Market growth (increased market size, increased sales)
    - Stealing share (same market size, increased sales)
    - Forward buying (same market size, same sales)
  - Four key factors influencing the timing of a promotion (peak season or off-season)
    - Impact of the promotion on demand
    - Cost of holding inventory
    - Cost of changing the level of capacity
    - Product margins



## 9.1. Responding to Predictable Variability in the Supply Chain

- Summary of Impact on Promoting Timing

| Factor                                  | Impact on Timing of Promotion/Forward Buy          |
|-----------------------------------------|----------------------------------------------------|
| High forward buying                     | Favors promotion during <b>low-demand periods</b>  |
| High ability steal market share         | Favors promotion during <b>peak-demand periods</b> |
| High ability to increase overall market | Favors promotion during <b>peak-demand periods</b> |
| High margin                             | Favors promotion during <b>peak-demand periods</b> |
| Low margin                              | Favors promotion during <b>low-demand periods</b>  |
| High manufacturer holding costs         | Favors promotion during <b>low-demand periods</b>  |
| High costs of changing capacity         | Favors promotion during <b>low-demand periods</b>  |
| High retailer holding costs             | Decreases forward buying by retailer               |
| High promotion elasticity of consumer   | Decreases forward buying by retailer               |



## 9.2. Sales and Operations Planning at Red Tomato

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- Pricing (or promotion) and aggregate production planning must be done jointly to maximize supply chain profitability, along with collaborative forecasting
  - so-called ‘collaborative sales and operations planning’
- Optimal split of supply chain profits achieved between two different members of the supply chain
- Example: Red Tomato Tools (manufacturer) & Green Thumb Gardens (retailer) – Collaborative sales and operations planning



## 9.2. Sales and Operations Planning at Red Tomato

- The base case

| Month    | Demand Forecast |
|----------|-----------------|
| January  | 1,600           |
| February | 3,000           |
| March    | 3,200           |
| April    | 3,800           |
| May      | 2,200           |
| June     | 2,200           |

- Retail price = \$40/unit
- Workforce in January = 80 workers (20 working days in each month)
- Starting inventory in January = 1,000 tools
- At least 500 units in inventory at the end of June
- Labor rule: No more than 10 hours of overtime/worker per month



## 9.2. Sales and Operations Planning at Red Tomato

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- Costs for Red Tomato and Green Thumb

| Item                              | Cost           |
|-----------------------------------|----------------|
| Material cost                     | \$10/unit      |
| Inventory holding cost            | \$2/unit/month |
| Marginal cost of stockout/backlog | \$5/unit/month |
| Hiring and training costs         | \$300/worker   |
| Layoff cost                       | \$500/worker   |
| Labor hours required              | 4/unit         |
| Regular time cost                 | \$4/hour       |
| Overtime cost                     | \$6/hour       |
| Cost of subcontracting            | \$30/unit      |



## 9.2. Sales and Operations Planning at Red Tomato

- Base case aggregate plan for Red Tomato and Green Thumb (table 8.3)

|    | A                                        | B              | C                 | D                     | E               | F                 | G                  | H                  | I                 | J             | K            |
|----|------------------------------------------|----------------|-------------------|-----------------------|-----------------|-------------------|--------------------|--------------------|-------------------|---------------|--------------|
| 1  | <b>Aggregate Plan Decision Variables</b> |                |                   |                       |                 |                   |                    |                    |                   |               |              |
| 2  |                                          | H <sub>t</sub> | L <sub>t</sub>    | W <sub>t</sub>        | O <sub>t</sub>  | I <sub>t</sub>    | S <sub>t</sub>     | C <sub>t</sub>     | P <sub>t</sub>    |               |              |
| 3  | <b>Period</b>                            | <b># Hired</b> | <b># Laid off</b> | <b># Workforce</b>    | <b>Overtime</b> | <b>Inventory</b>  | <b>Stockout</b>    | <b>Subcontract</b> | <b>Production</b> | <b>Demand</b> | <b>Price</b> |
| 4  | 0                                        | 0              | 0                 | 80                    | 0               | 1,000             | 0                  | 0                  |                   |               |              |
| 5  | 1                                        | 0              | 16                | 64                    | 0               | 1,960             | 0                  | 0                  | 2,560             | 1,600         | 40           |
| 6  | 2                                        | 0              | 0                 | 64                    | 0               | 1,520             | 0                  | 0                  | 2,560             | 3,000         | 40           |
| 7  | 3                                        | 0              | 0                 | 64                    | 0               | 880               | 0                  | 0                  | 2,560             | 3,200         | 40           |
| 8  | 4                                        | 0              | 0                 | 64                    | 0               | 0                 | 220                | 140                | 2,560             | 3,800         | 40           |
| 9  | 5                                        | 0              | 0                 | 64                    | 0               | 140               | 0                  | 0                  | 2,560             | 2,200         | 40           |
| 10 | 6                                        | 0              | 0                 | 64                    | 0               | 500               | 0                  | 0                  | 2,560             | 2,200         | 40           |
| 22 | <b>Total Cost =</b>                      |                | <b>\$422,660</b>  |                       |                 |                   |                    |                    |                   |               |              |
| 23 |                                          |                |                   |                       |                 | <b>Base Price</b> |                    | <b>\$ 40</b>       |                   |               |              |
| 24 | <b>Total Revenue =</b>                   |                | <b>\$640,000</b>  | <b>Promote? (0/1)</b> |                 | <b>0</b>          | <b>Consumption</b> |                    | <b>0.10</b>       |               |              |
| 25 | <b>Profit =</b>                          |                | <b>\$217,340</b>  | <b>Month (1/4)</b>    |                 | <b>1</b>          | <b>Forward buy</b> |                    | <b>0.20</b>       |               |              |

- Total cost over planning horizon = \$422,660
- Revenue over planning horizon = \$640,000
- Profit over planning horizon = \$217,340



## 9.2. Sales and Operations Planning at Red Tomato

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- When to promote: Peak of Off-Peak
  - Scheme: Retail sale price \$40 → \$39 in one month
  - Question: **When**? January (off-peak month) or April (peak month)?
  - Estimation by analysis: 20% of each of the two following months' demand is moved forward
- Scenarios
  - 1) January promotion: 10% increase in consumption in January
  - 2) April promotion: 10% increase in consumption in April
  - 3) January promotion: 100% increase in consumption in January
  - 4) April promotion: 100% increase in consumption in April
  - More: When the regular retail price is lowered, \$40 → \$31



## 9.2. Sales and Operations Planning at Red Tomato

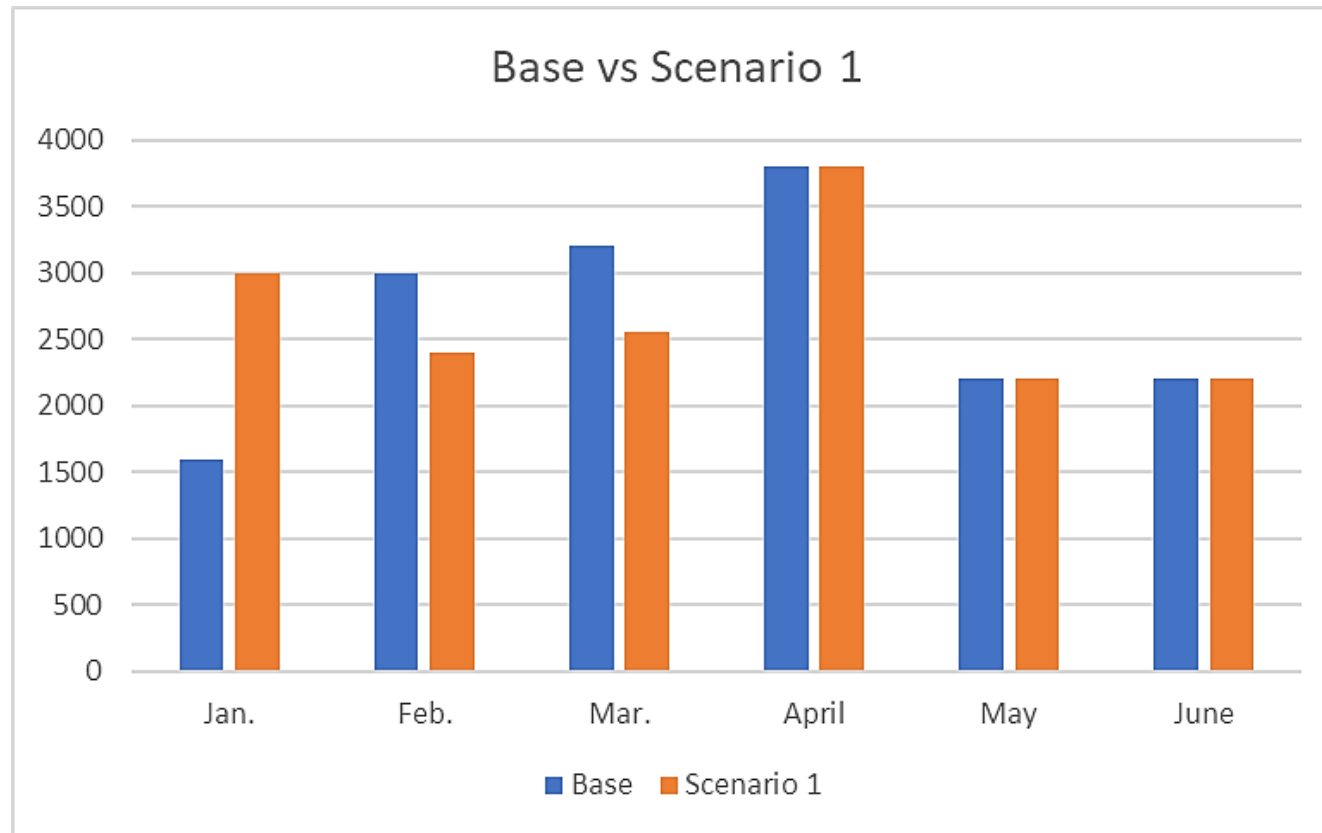
- Scenario 1: January discount
  - Price = \$39, 10% Increase, 20% Forward Buy

| Month                          | Demand Forecast             |           |
|--------------------------------|-----------------------------|-----------|
|                                | Jan. Discount               | Base Case |
| Jan. (10% increase)            | 3,000<br>(1600+160+600+640) | 1,600     |
| Feb. (20% forward buy in Jan.) | 2,400<br>(3000*0.8)         | 3,000     |
| Mar. (20% forward buy in Jan.) | 2,560<br>(3200*0.8)         | 3,200     |
| April                          | 3,800                       | 3,800     |
| May                            | 2,200                       | 2,200     |
| June                           | 2,200                       | 2,200     |
| Sum                            | 16,160                      | 16,000    |



## 9.2. Sales and Operations Planning at Red Tomato

- Scenario 1: January discount
  - Price = \$39, 10% Increase, 20% Forward Buy



## 9.2. Sales and Operations Planning at Red Tomato

- Optimal Aggregate Plan for January Promotion (10% Increase)

|    | A                                 | B              | C              | D              | E              | F              | G              | H              | I              | J      | K     |
|----|-----------------------------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|--------|-------|
| 1  | Aggregate Plan Decision Variables |                |                |                |                |                |                |                |                |        |       |
| 2  |                                   | H <sub>t</sub> | L <sub>t</sub> | W <sub>t</sub> | O <sub>t</sub> | I <sub>t</sub> | S <sub>t</sub> | C <sub>t</sub> | P <sub>t</sub> |        |       |
| 3  | Period                            | # Hired        | # Laid off     | # Workforce    | Overtime       | Inventory      | Stockout       | Subcontract    | Production     | Demand | Price |
| 4  | 0                                 | 0              | 0              | 80             | 0              | 1,000          | 0              | 0              |                |        |       |
| 5  | 1                                 | 0              | 15             | 65             | 0              | 600            | 0              | 0              | 2,600          | 3,000  | 39    |
| 6  | 2                                 | 0              | 0              | 65             | 0              | 800            | 0              | 0              | 2,600          | 2,400  | 40    |
| 7  | 3                                 | 0              | 0              | 65             | 0              | 840            | 0              | 0              | 2,600          | 2,560  | 40    |
| 8  | 4                                 | 0              | 0              | 65             | 0              | 0              | 300            | 60             | 2,600          | 3,800  | 40    |
| 9  | 5                                 | 0              | 0              | 65             | 0              | 100            | 0              | 0              | 2,600          | 2,200  | 40    |
| 10 | 6                                 | 0              | 0              | 65             | 0              | 500            | 0              | 0              | 2,600          | 2,200  | 40    |
| 22 | Total Cost =                      |                | \$422,080      |                |                |                |                |                |                |        |       |
| 23 |                                   |                |                |                |                | Base Price     |                | \$40           |                |        |       |
| 24 | Total Revenue =                   |                | \$643,400      | Promote? (0/1) |                | 1              | Consumption    |                | 0.10           |        |       |
| 25 | Profit =                          |                | \$221,320      | Month (1/4)    |                | 1              | Forward buy    |                | 0.20           |        |       |

- Total cost over planning horizon = \$422,080 (<\$422,660)  
Revenue over planning horizon = \$643,400 (>\$640,000)  
Profit over planning horizon = \$221,320 (> \$217,340)
- Lower seasonal inventory, Somewhat lower total cost, Higher total profit



## 9.2. Sales and Operations Planning at Red Tomato

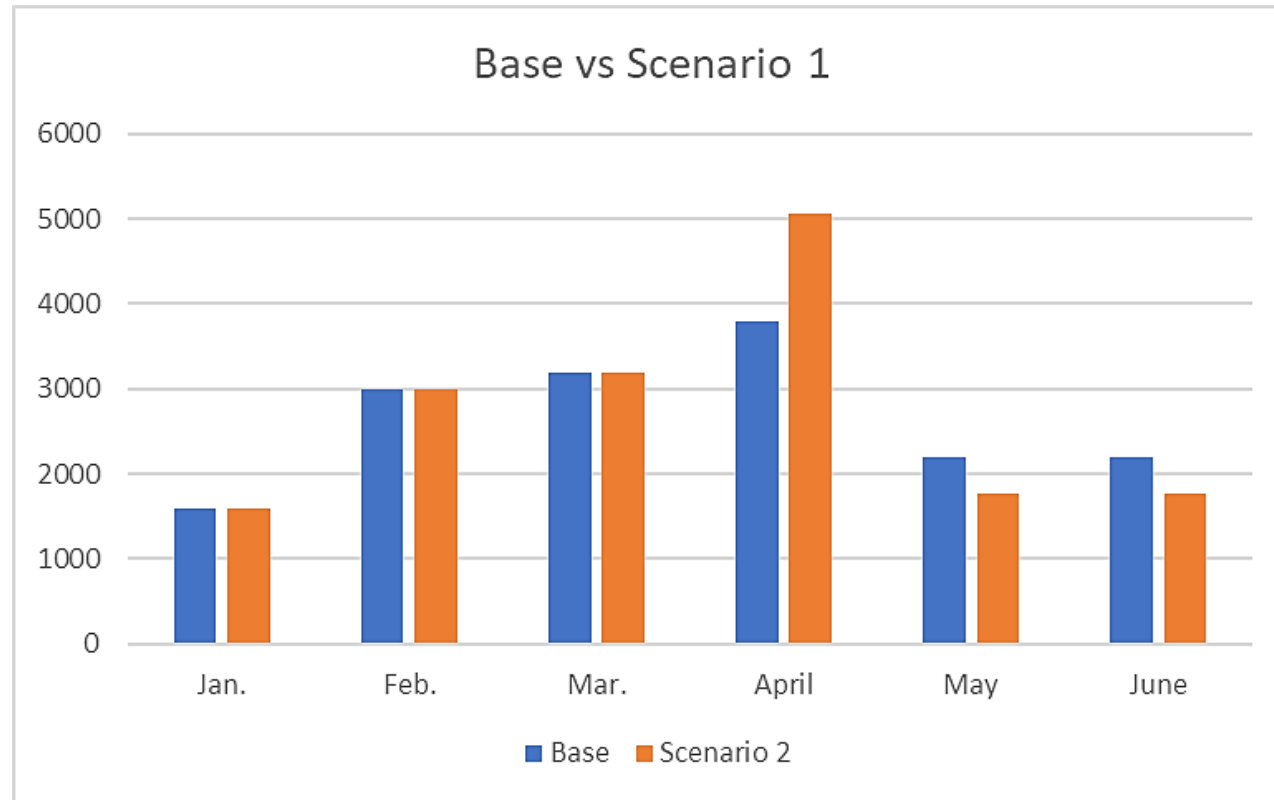
- Scenario 2: April discount
  - Price = \$39, 10% Increase, 20% Forward Buy

| Month                           | Demand Forecast |           |
|---------------------------------|-----------------|-----------|
|                                 | April Discount  | Base Case |
| Jan.                            | 1,600           | 1,600     |
| Feb.                            | 3,000           | 3,000     |
| Mar.                            | 3,200           | 3,200     |
| April (10% increase)            | 5,060           | 3,800     |
| May (20% forward buy in April)  | 1,760           | 2,200     |
| June (20% forward buy in April) | 1,760           | 2,200     |
| Sum                             | 16,380          | 16,000    |



## 9.2. Sales and Operations Planning at Red Tomato

- Scenario 2: April discount
  - Price = \$39, 10% Increase, 20% Forward Buy



## 9.2. Sales and Operations Planning at Red Tomato

- Optimal Aggregate Plan for April Promotion (10% Increase)

|    | A                                 | B              | C              | D              | E              | F                | G              | H              | I              | J      | K     |
|----|-----------------------------------|----------------|----------------|----------------|----------------|------------------|----------------|----------------|----------------|--------|-------|
| 1  | Aggregate Plan Decision Variables |                |                |                |                |                  |                |                |                |        |       |
| 2  |                                   | H <sub>t</sub> | L <sub>t</sub> | W <sub>t</sub> | O <sub>t</sub> | I <sub>t</sub>   | S <sub>t</sub> | C <sub>t</sub> | P <sub>t</sub> |        |       |
| 3  | Period                            | # Hired        | # Laid off     | # Workforce    | Overtime       | Inventory        | Stockout       | Subcontract    | Production     | Demand | Price |
| 4  | 0                                 | 0              | 0              | 80             | 0              | 1,000            | 0              | 0              |                |        |       |
| 5  | 1                                 | 0              | 14             | 66             | 0              | 2,040            | 0              | 0              | 2,640          | 1,600  | 40    |
| 6  | 2                                 | 0              | 0              | 66             | 0              | 1,680            | 0              | 0              | 2,640          | 3,000  | 40    |
| 7  | 3                                 | 0              | 0              | 66             | 0              | 1,120            | 0              | 0              | 2,640          | 3,200  | 40    |
| 8  | 4                                 | 0              | 0              | 66             | 0              | 0                | 1,260          | 40             | 2,640          | 5,060  | 39    |
| 9  | 5                                 | 0              | 0              | 66             | 0              | 0                | 380            | 0              | 2,640          | 1,760  | 40    |
| 10 | 6                                 | 0              | 0              | 66             | 0              | 500              | 0              | 0              | 2,640          | 1,760  | 40    |
| 22 | Total Cost =                      |                | \$ 438,920     |                |                |                  |                |                |                |        |       |
| 23 |                                   |                |                |                |                | Base Price \$ 40 |                |                |                |        |       |
| 24 | Total Revenue =                   | \$ 650,140     | Promote? (0/1) |                | 1              | Consumption      |                | 0.10           |                |        |       |
| 25 | Profit =                          | \$ 211,220     | Month (1/4)    |                | 4              | Forward buy      |                | 0.20           |                |        |       |

- Total cost over planning horizon = \$438,920 (> \$422,660)  
Revenue over planning horizon = \$650,140 (> \$640,000)  
Profit over planning horizon = \$211,220 (< \$217,340), (< \$221,320 for Jan. Promotion)
- Higher seasonal inventory, Somewhat higher total cost, Slightly smaller total profit



## 9.2. Sales and Operations Planning at Red Tomato

- Scenario 3: January discount
  - Price = \$39, 100% Increase, 20% Forward Buy

| Month                          | Demand Forecast |           |
|--------------------------------|-----------------|-----------|
|                                | Jan. Discount   | Base Case |
| Jan. (100% increase)           | 4,440           | 1,600     |
| Feb. (20% forward buy in Jan.) | 2,400           | 3,000     |
| Mar. (20% forward buy in Jan.) | 2,560           | 3,200     |
| April                          | 3,800           | 3,800     |
| May                            | 2,200           | 2,200     |
| June                           | 2,200           | 2,200     |
| Sum                            | 17,600          | 16,000    |



## 9.2. Sales and Operations Planning at Red Tomato

- Optimal Aggregate Plan for January Promotion (100% Increase)

|    | A                                 | B              | C              | D              | E              | F              | G              | H              | I              | J      | K     |  |
|----|-----------------------------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|--------|-------|--|
| 1  | Aggregate Plan Decision Variables |                |                |                |                |                |                |                |                |        |       |  |
| 2  |                                   | H <sub>t</sub> | L <sub>t</sub> | W <sub>t</sub> | O <sub>t</sub> | I <sub>t</sub> | S <sub>t</sub> | C <sub>t</sub> | P <sub>t</sub> |        |       |  |
| 3  | Period                            | # Hired        | # Laid off     | # Workforce    | Overtime       | Inventory      | Stockout       | Subcontract    | Production     | Demand | Price |  |
| 4  | 0                                 | 0              | 0              | 80             | 0              | 1,000          | 0              | 0              |                |        |       |  |
| 5  | 1                                 | 0              | 0              | 80             | 0              | 0              | 140            | 100            | 3,200          | 4,440  | 39    |  |
| 6  | 2                                 | 0              | 11             | 69             | 0              | 220            | 0              | 0              | 2,760          | 2,400  | 40    |  |
| 7  | 3                                 | 0              | 0              | 69             | 0              | 420            | 0              | 0              | 2,760          | 2,560  | 40    |  |
| 8  | 4                                 | 0              | 0              | 69             | 0              | 0              | 620            | 0              | 2,760          | 3,800  | 40    |  |
| 9  | 5                                 | 0              | 0              | 69             | 0              | 0              | 60             | 0              | 2,760          | 2,200  | 40    |  |
| 10 | 6                                 | 0              | 0              | 69             | 0              | 500            | 0              | 0              | 2,760          | 2,200  | 40    |  |
| 22 | Total Cost =                      |                | \$ 456,880     |                |                |                |                |                |                |        |       |  |
| 23 |                                   |                |                |                |                | Base Price     |                | \$ 40          |                |        |       |  |
| 24 | Total Revenue =                   |                | \$ 699,560     | Promote? (0/1) |                | 1              | Consumption    |                | 1.00           |        |       |  |
| 25 | Profit =                          |                | \$ 242,680     | Month (1/4)    |                | 1              | Forward buy    |                | 0.20           |        |       |  |

- Total cost over planning horizon = \$456,880 (> \$422,660, \$34,220)
- Revenue over planning horizon = \$699,560 (> \$640,000, \$59,560)
- Profit over planning horizon = \$242,680 (> \$217,340, \$25,340)
- Higher total profit than base case
- Split of profit \$25,340 (\$59,560 - \$34,220)



## 9.2. Sales and Operations Planning at Red Tomato

- Scenario 4: April discount
  - Price = \$39, 100% Increase, 20% Forward Buy

| Month                           | Demand Forecast |           |
|---------------------------------|-----------------|-----------|
|                                 | April Discount  | Base Case |
| Jan.                            | 1,600           | 1,600     |
| Feb.                            | 3,000           | 3,000     |
| Mar.                            | 3,200           | 3,200     |
| April (100% increase)           | 8,480           | 3,800     |
| May (20% forward buy in April)  | 1,760           | 2,200     |
| June (20% forward buy in April) | 1,760           | 2,200     |
| Sum                             | 19,800          | 16,000    |



## 9.2. Sales and Operations Planning at Red Tomato

- Optimal Aggregate Plan for April Promotion (100% Increase)

|    | A                                        | B              | C                 | D                  | E               | F                | G               | H                  | I                 | J             | K            |
|----|------------------------------------------|----------------|-------------------|--------------------|-----------------|------------------|-----------------|--------------------|-------------------|---------------|--------------|
| 1  | <b>Aggregate Plan Decision Variables</b> |                |                   |                    |                 |                  |                 |                    |                   |               |              |
| 2  |                                          | H <sub>t</sub> | L <sub>t</sub>    | W <sub>t</sub>     | O <sub>t</sub>  | I <sub>t</sub>   | S <sub>t</sub>  | C <sub>t</sub>     | P <sub>t</sub>    |               |              |
| 3  | <b>Period</b>                            | <b># Hired</b> | <b># Laid off</b> | <b># Workforce</b> | <b>Overtime</b> | <b>Inventory</b> | <b>Stockout</b> | <b>Subcontract</b> | <b>Production</b> | <b>Demand</b> | <b>Price</b> |
| 4  | 0                                        | 0              | 0                 | 80                 | 0               | 1,000            | 0               | 0                  |                   |               |              |
| 5  | 1                                        | 0              | 0                 | 80                 | 0               | 2,600            | 0               | 0                  | 3,200             | 1,600         | 40           |
| 6  | 2                                        | 0              | 0                 | 80                 | 0               | 2,800            | 0               | 0                  | 3,200             | 3,000         | 40           |
| 7  | 3                                        | 0              | 0                 | 80                 | 0               | 2,800            | 0               | 0                  | 3,200             | 3,200         | 40           |
| 8  | 4                                        | 0              | 0                 | 80                 | 0               | 0                | 2,380           | 100                | 3,200             | 8,480         | 39           |
| 9  | 5                                        | 0              | 0                 | 80                 | 0               | 0                | 940             | 0                  | 3,200             | 1,760         | 40           |
| 10 | 6                                        | 0              | 0                 | 80                 | 0               | 500              | 0               | 0                  | 3,200             | 1,760         | 40           |
| 22 | <b>Total Cost =</b>                      |                | <b>\$ 536,200</b> |                    |                 |                  |                 |                    |                   |               |              |
| 23 |                                          |                |                   |                    |                 |                  |                 |                    |                   |               |              |
| 24 | <b>Total Revenue =</b>                   | \$ 783,520     |                   | Promote? (0/1)     | 1               | Base Price       |                 | \$ 40              |                   |               |              |
| 25 | <b>Profit =</b>                          | \$ 247,320     |                   | Month (1/4)        | 4               | Consumption      |                 | 1.00               |                   |               |              |
|    |                                          |                |                   |                    |                 | Forward buy      |                 | 0.20               |                   |               |              |

- Total cost over planning horizon = \$536,200 (> \$456,880 for Scenario 3)  
Revenue over planning horizon = \$783,520 (> \$699,560 for Scenario 3)  
Profit over planning horizon = **\$247,320** (> \$242,680 for Scenario 3)
- Much higher level of seasonal inventory, Uses more stockouts and subcontracting
- Revenues increase, Higher profit



## 9.2. Sales and Operations Planning at Red Tomato

- Supply chain performance under different scenarios

| Scenario  | Regular Price | Promotion Price | Promotion Period | Percentage of Increase in Demand | Percentage of Forward Buy | Profit           | Average Inventory |
|-----------|---------------|-----------------|------------------|----------------------------------|---------------------------|------------------|-------------------|
| Base Case | \$40          | \$40            | NA               | NA                               | NA                        | \$217,340        | 875               |
| 1         | \$40          | \$39            | January          | 10%                              | 20%                       | <b>\$221,320</b> | 515               |
| 2         | \$40          | \$39            | April            | 10%                              | 20%                       | \$211,220        | 932               |
| 3         | \$40          | \$39            | January          | 100%                             | 20%                       | \$242,680        | 232               |
| 4         | \$40          | \$39            | April            | 100%                             | 20%                       | <b>\$247,320</b> | 1,492             |
| 5         | \$31          | \$31            | NA               | NA                               | NA                        | \$73,340         | 875               |
| 6         | \$31          | \$30            | January          | 100%                             | 20%                       | <b>\$84,280</b>  | 232               |
| 7         | \$31          | \$30            | April            | 100%                             | 20%                       | \$69,120         | 1,492             |



## 9.2. Sales and Operations Planning at Red Tomato

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- Conclusions regarding the impact of promotions
  - Average inventory increases **if** a promotion is run during the peak period and decreases if a promotion is run during the off-peak period
  - Promoting during a peak period may decrease overall profitability **if** there is a small increase in consumption and a significant fraction of a demand increase results from a forward buy
  - **As** the consumption increase from discounting grows and forward buying becomes a smaller fraction of the demand increase from a promotion, it is more profitable to promote during the peak period
  - **As** the product margin declines, promoting during the peak- demand period becomes less profitable



## 9.2. Sales and Operations Planning at Red Tomato

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- Essay Question

- 1) Discuss how a firm can respond to predictable variability of demand in the supply chain.
- 2) Discuss the importance of collaboration within a supply chain when performing aggregate planning.



## 9.2. Sales and Operations Planning at Red Tomato

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- Discuss how a firm can respond to predictable variability of demand in the supply chain.
- **Answer:** Faced with predictable variability, a company's goal is to respond in a manner that maximizes profitability. A firm must choose how to handle predictable variability by utilizing techniques in two broad categories:
  - **Manage supply using capacity, inventory, subcontracting, and backlogs.**
  - **Manage demand using short-term price discounts and trade promotions.**
- The use of these tools enables the supply chain to greatly increase its profitability because it is able to match supply and demand in a much more coordinated fashion.
- One way to meet seasonal demand requires carrying enough manufacturing capacity to meet demand in any period. The advantage of this approach is very low inventory costs, because no inventory needs to be carried from period to period. The disadvantage, however, is that much of the expensive capacity would go unused during most months when demand was lower. Another approach to meeting demand would be to build up inventory during the off season to keep production stable year round. The advantage of this approach lies in the fact that a firm could get by with a smaller, less expensive factory. High inventory carrying costs, however, make this alternative expensive. A third approach would be for a firm to work with their retail partners in the supply chain to offer a price promotion during periods of low demand. This promotion shifts some of the demand into a slow period, thereby spreading demand more evenly throughout the year and reducing the seasonal surge. Such a demand pattern is less expensive to supply. A company needs to decide which alternative maximizes their profitability.



## 9.2. Sales and Operations Planning at Red Tomato

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- Discuss the importance of collaboration within a supply chain when performing aggregate planning.
- **Answer:** Supply chains can influence demand by using pricing and other forms of promotion. Marketing and sales often make the promotion and pricing decisions, and they typically make them with the objective of maximizing revenue. However, changing the demand pattern can change the cost the company incurs to meet that demand. Thus, pricing decisions based only on revenue considerations often result in a decrease in overall profitability. The same is true when thinking of the supply chain. The retailer sets the price and runs promotions to generate demand. This is regularly done without taking into account the impact on the rest of the supply chain. Therefore, the combination of pricing and aggregate planning (both demand and supply management) can be used to maximize supply chain profitability.
- When performing aggregate planning, the goal of all firms in the supply chain should be to maximize supply chain profits because this outcome leaves them more to divide with each other. For profit maximization to take place, companies need to devise a way to collaborate and, just as important, determine a way to split the supply chain profits.
- Determining how these profits will be allocated to different members of the supply chain is a key to successful collaboration.

